

Voice Controlled Toy Cybertruck - Technical Report

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1. Abstract

The Voice Controlled Toy Cybertruck is a voice-operated robotic vehicle built using an Arduino Uno, L293D motor driver shield, HC-05 and HC-06 Bluetooth modules, VC-02 offline voice recognition module and four BO motors. The remote recognizes five predefined voice commands and transmits hexadecimal data wirelessly to the vehicle. The Arduino receives the data through the HC-06 Bluetooth module and controls the motors accordingly. The robot is powered by Li-ion batteries and enclosed in a Cybertruck-inspired cardboard body. This report explains the hardware, software, construction and working of the complete project so that another student can understand and recreate it.

2. Introduction

Robotics combines electronics, programming and mechanical design to perform useful tasks automatically. Voice-controlled robots are an interesting application because they allow a user to control a machine through simple spoken commands. In this project, an offline voice recognition module was used so that the system can work without a mobile phone or internet connection. The project consists of a handheld remote and a robotic vehicle communicating through Bluetooth. The design of the outer body was inspired by the Tesla Cybertruck to give the project a modern appearance suitable for exhibitions. The project also helped in understanding Bluetooth communication, Arduino programming, motor control and embedded system integration.

3. Objectives

- Develop a voice-controlled robotic vehicle.
- Implement offline voice recognition.
- Establish Bluetooth communication.
- Control four BO motors using Arduino Uno and L293D.
- Design a Cybertruck-inspired body.
- Create a portable handheld remote.
- Demonstrate Forward, Backward, Left, Right and Stop commands.
- Document the complete project.

4. Project Overview

The project consists of two units: a remote controller and the robotic vehicle. The user speaks one of the five predefined commands into the microphone connected to the VC-02 voice recognition module. After recognizing the command, the VC-02 provides voice feedback and sends the corresponding hexadecimal data through the HC-05 Bluetooth module. The HC-06 module mounted on the vehicle receives the data and forwards it to the Arduino Uno. The Arduino processes the received command and drives the motors through the L293D motor driver shield. The vehicle continues moving until another valid command is received. The complete system is powered using rechargeable Li-ion batteries and enclosed in a cardboard body inspired by the Tesla Cybertruck.



Figure 4.1: final image of the completed Toy Cybertruck.



Figure 4.2: final image of the remote controller.

5. Hardware Components

5.1 Arduino Uno

The Arduino Uno is the main controller of the Voice Controlled Toy Cybertruck. It receives the hexadecimal commands from the HC-06 Bluetooth module, processes them and controls the L293D motor driver shield to move the motors.

Specification	Value
Microcontroller	ATmega328P
Operating Voltage	5 V
Digital I/O Pins	14
PWM Pins	6
Flash Memory	32 KB

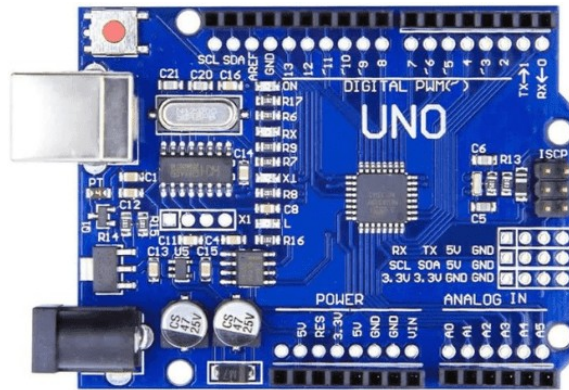


Figure 5.1: Arduino Uno

5.2 L293D Motor Driver Shield

The L293D motor driver shield acts as an interface between the Arduino Uno and the DC motors. Since an Arduino cannot supply sufficient current to drive motors directly, the shield amplifies the control signals and powers the four BO motors.

Specification	Value
Motor Channels	4 DC Motors
Logic Voltage	5 V
Motor Supply	Up to 12 V
Purpose	Motor Driving

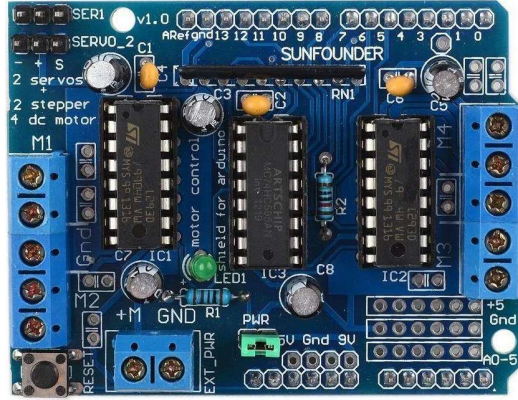


Figure 5.2: L293D Motor Driver Shield.

5.3 HC-05 Bluetooth Module

The HC-05 module is installed inside the handheld remote. It wirelessly transmits the hexadecimal commands generated by the VC-02 voice recognition module to the HC-06 module mounted on the vehicle.

Specification	Value
Mode	Master
Communication	UART
Frequency	2.4 GHz
Range	Approx. 10 m

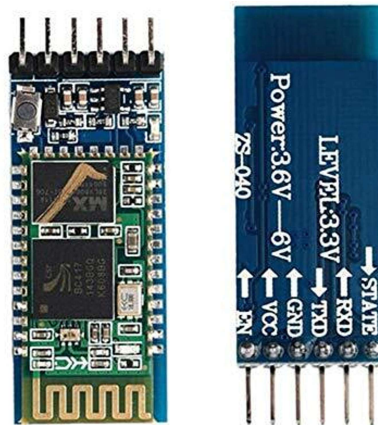


Figure 5.3: HC-05 Bluetooth Module.

5.4 HC-06 Bluetooth Module

The HC-06 Bluetooth module is installed on the robotic vehicle. It receives the data sent by the HC-05 module and forwards it to the Arduino Uno through serial communication.

Specification	Value
Mode	Slave
Communication	UART
Frequency	2.4 GHz
Range	Approx. 10 m



Figure 5.4: HC-06 Bluetooth Module.

5.5 VC-02 Voice Recognition Module

The VC-02 module performs offline voice recognition. It recognizes the predefined commands spoken by the user, gives audio feedback through the connected speaker and sends the corresponding hexadecimal data to the HC-05 Bluetooth module.

Specification	Value
Recognition	Offline
Input	Microphone
Output	Speaker + UART

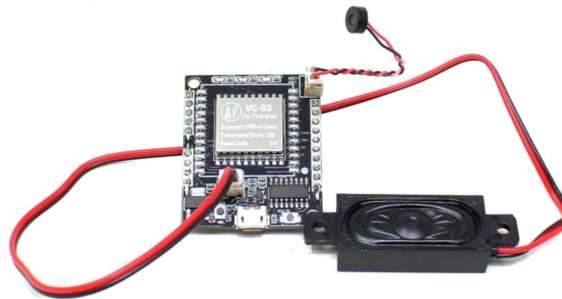


Figure 5.5: VC-02 Voice Recognition Module.

5.6 BO Motors and Wheels

The vehicle uses four BO geared DC motors with four wheels. The L293D shield controls the direction of rotation of each side of the vehicle, allowing forward, backward and turning movements.

Specification	Value
Quantity	4
Type	BO Geared DC Motors
Drive	4 Wheel



Figure5.6 : BO Motor and Wheel.

5.7 Li-ion Batteries

Two rechargeable 3.7 V 3000 mAh Li-ion cells connected in series power the robotic vehicle. A separate 3.7 V 2200 mAh Li-ion battery powers the handheld remote.

Specification	Value
Vehicle Battery	$2 \times 3.7 \text{ V } 3000 \text{ mAh}$
Remote Battery	$1 \times 3.7 \text{ V } 2200 \text{ mAh}$



Figure 5.7: Li-ion Batteries.

5.8 Rocker Switch

A 10 A rocker switch is used in both the vehicle and the remote to safely switch the power supply ON and OFF.

Specification	Value
Type	Rocker Switch
Rating	10 A



Figure 5.8: Rocker Switch.

5.9 Chassis

The chassis forms the mechanical base of the project. It supports the motors, battery holder, Arduino, Bluetooth module and enclosure while providing rigidity to the complete system.

Specification	Value
Material	Robot Car Chassis
Purpose	Mechanical Support



Figure 5.9: Chassis.

5.10 Cardboard Enclosure

A custom enclosure was designed using cardboard and inspired by the Tesla Cybertruck. The enclosure protects the electronics, improves appearance and allows easy access for maintenance.

Specification	Value
Material	Cardboard
Design	Cybertruck Inspired
Feature	Removable Cover

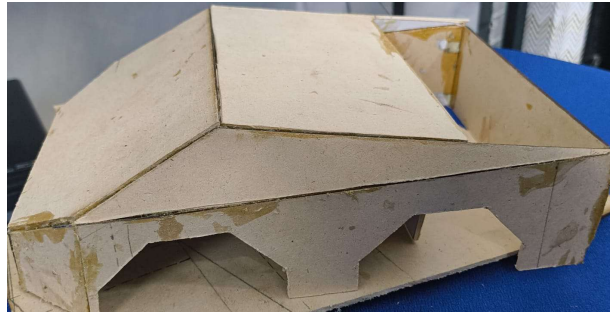


Figure 5.10: Cardboard Enclosure.

6. Software Used

The software for the Voice Controlled Toy Cybertruck was developed using Arduino IDE Version 2.3.7. The complete program was written in C++ and uploaded to the Arduino Uno using a USB cable.

Software/Library	Version	Purpose
Arduino IDE	2.3.7	Writing and uploading the program
AFMotor.h	Library	Control of L293D motor shield
SoftwareSerial.h	Library	Serial communication

7. Mechanical Design

The mechanical structure of the project consists of a four-wheel robot chassis covered with a custom Cybertruck-inspired cardboard body. The enclosure protects the electronic components while improving the appearance of the vehicle during demonstrations. The cover was designed so that it can be removed whenever maintenance is required. The switch remains accessible from outside the body and sufficient space is provided inside for the Arduino Uno, L293D shield, HC-06 module and battery holder.

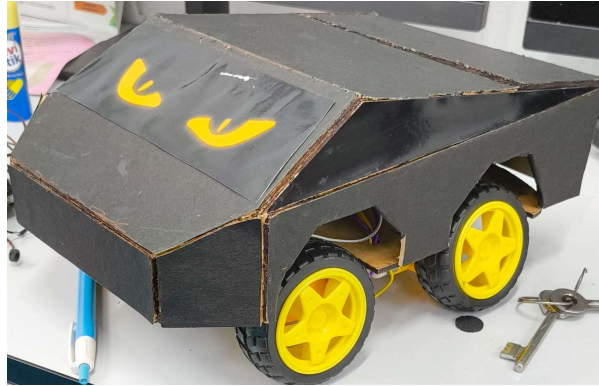


Figure 7.1: Completed Toy Cybertruck.

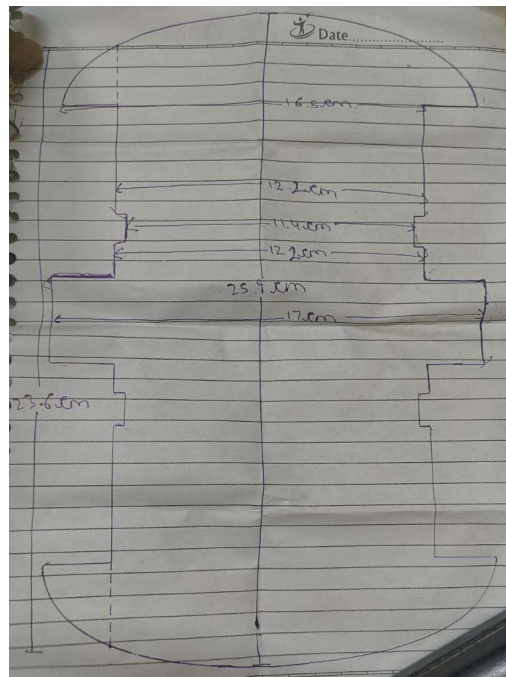


Figure 7.2: Chassis dimension tracing sheet.

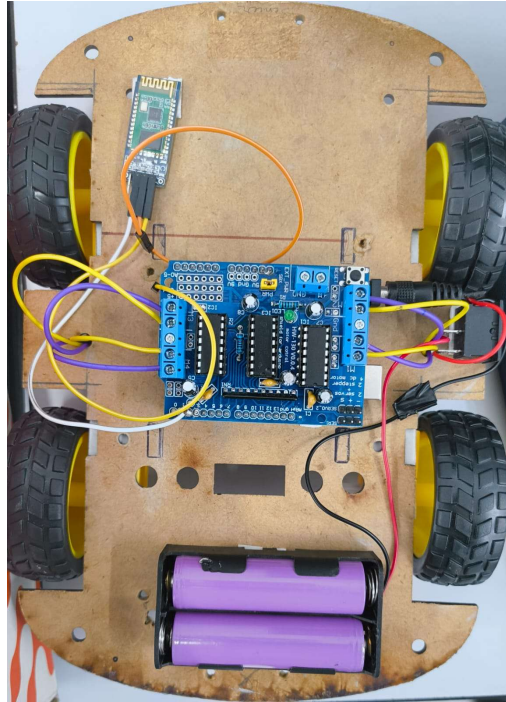


Figure 7.3: Internal layout of the car.

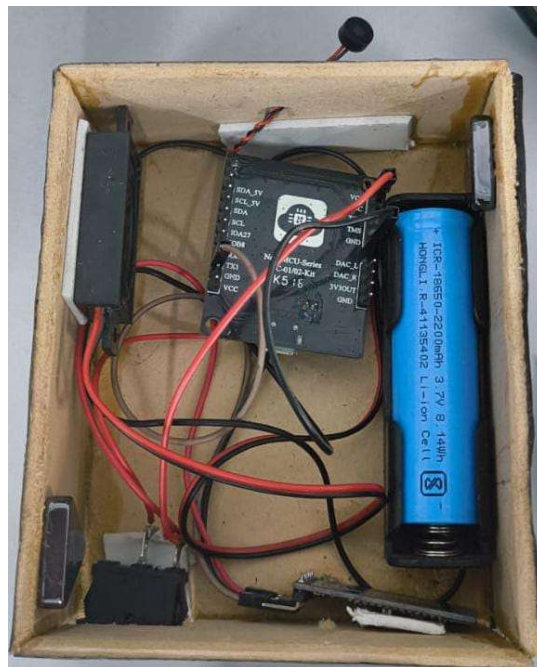


Figure 7.4: Internal layout of the remote.

Component Placement

- HC-06 Bluetooth module was placed inside the vehicle.
- Arduino Uno with L293D shield was mounted near the centre of the chassis.
- Battery holder was placed inside the enclosure.
- Rocker switch was kept accessible from outside.
- The remote contains the VC-02 module, HC-05 module, battery and speaker inside a cardboard enclosure.

8. Circuit Diagram

This chapter explains the electrical connections used in the project. The actual circuit diagrams will be inserted later. These diagrams should clearly show all wiring connections used in the final project.

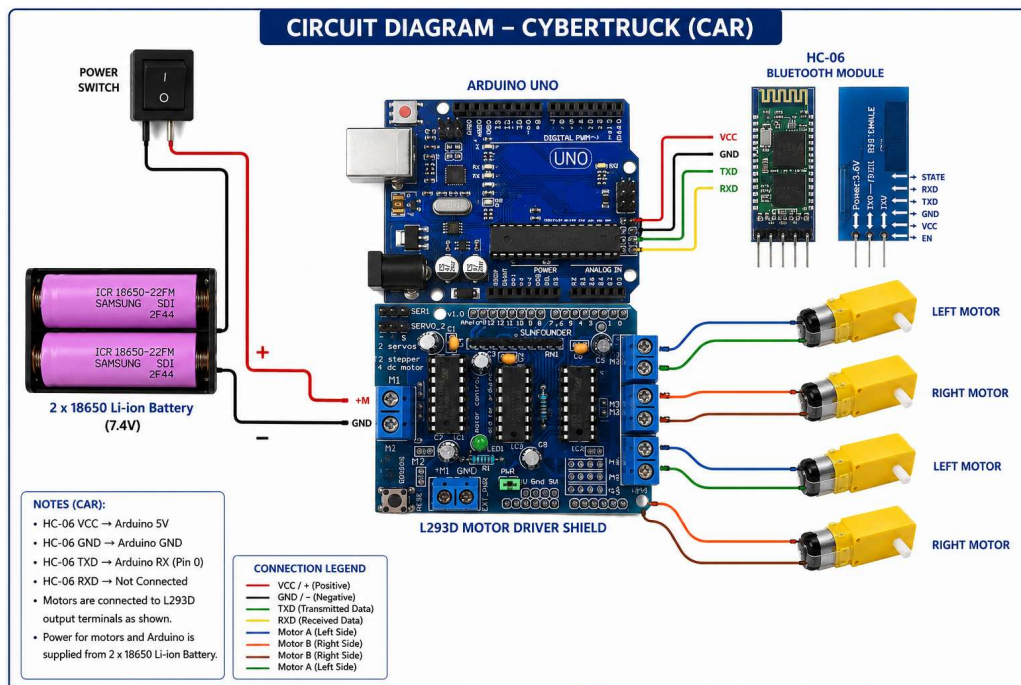


Figure 8.1: Circuit diagram of car.

Description:

- HC-06 TX connected to Arduino RX.
- HC-06 powered from the Arduino.
- L293D shield mounted on Arduino Uno.
- Four BO motors connected to the motor shield outputs.
- Two Li-ion batteries connected through the rocker switch.

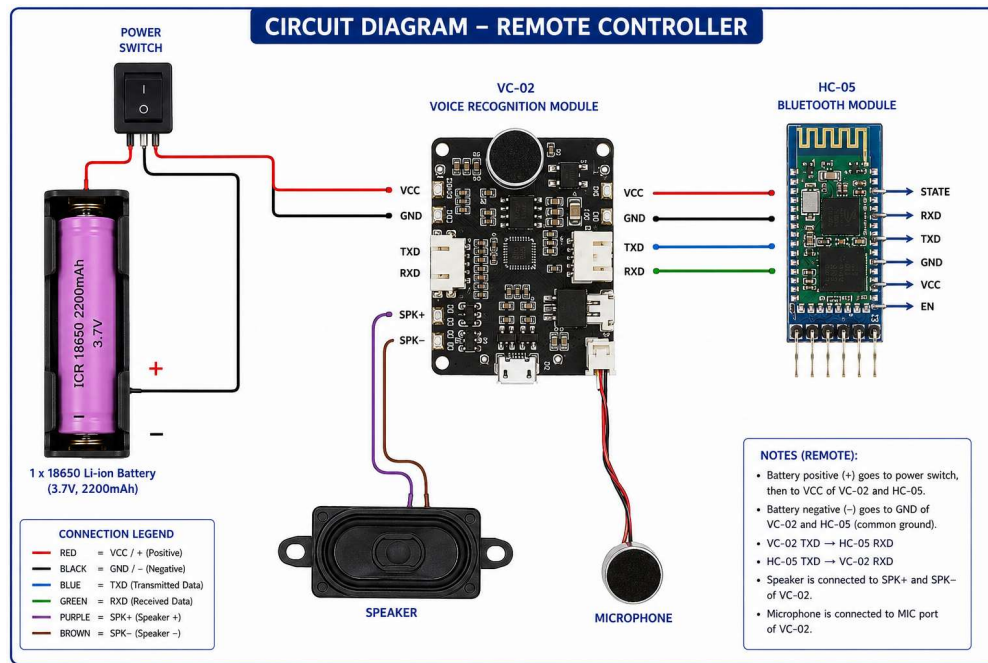


Figure 8.2: Circuit diagram of the handheld remote.

Description:

- Microphone connected to VC-02.
- Speaker connected to VC-02.
- VC-02 TX connected to HC-05 RX.
- HC-05 powered using a 3.7 V Li-ion battery through the rocker switch.

9. Working Principle

The Voice Controlled Toy Cybertruck operates through wireless communication between a handheld remote and the robotic vehicle. The complete operation begins when the user speaks one of the predefined commands such as Forward, Backward, Left, Right or Stop into the microphone connected to the VC-02 voice recognition module. The VC-02 has been trained to recognise these commands offline, therefore no internet connection is required.

After recognising the command, the VC-02 confirms the command through the connected speaker and generates a corresponding hexadecimal data packet. This hexadecimal value is sent to the HC-05 Bluetooth module present inside the remote. The HC-05 transmits the data wirelessly to the HC-06 Bluetooth module mounted on the robotic vehicle.

The HC-06 forwards the received data to the Arduino Uno through serial communication. The Arduino continuously checks for incoming Bluetooth data. Whenever a valid hexadecimal command is received, the Arduino compares it with the programmed command values and determines the required movement.

After identifying the command, the Arduino sends suitable control signals to the L293D Motor Driver Shield. The motor driver provides sufficient current to the four BO motors and rotates them in the required direction. The robot continues moving until a new valid command is received. For example, after receiving the Forward command, the vehicle keeps moving forward until the user issues another command such as Stop or Left.

Working Sequence:

User → Microphone → VC-02 → Speaker Feedback → HC-05 → HC-06 → Arduino Uno → L293D Motor Driver → BO Motors → Vehicle Movement

10. Algorithm

1. Start the system.
2. Switch ON both the remote and the robotic vehicle.
3. Wait until HC-05 and HC-06 establish Bluetooth communication.
4. Wait for the user to speak a valid command.
5. VC-02 recognises the command and generates the hexadecimal data.
6. Transmit the hexadecimal data through HC-05.
7. Receive the data using HC-06.
8. Arduino reads and processes the received command.
9. Drive the motors using the L293D motor driver shield.
10. Continue the current movement until another valid command is received.
11. Repeat the process until the system is switched OFF.

11. Assembly Procedure

The following steps were followed during the assembly of the project:

Step 1: Fix the four BO motors on the chassis using the motor brackets.

Step 2: Mount the Arduino Uno at the centre of the chassis and place the L293D Motor Driver Shield on it.

Step 3: Connect all four motors to the motor driver shield.

Step 4: Install the HC-06 Bluetooth module and complete its wiring with the Arduino.

Step 5: Connect the rocker switch and the battery holder to power the vehicle.

Step 6: Design and assemble the Cybertruck-inspired cardboard enclosure.

Step 7: Build the handheld remote using the VC-02 module, HC-05 module, microphone, speaker, battery and rocker switch.

Step 8: Upload the Arduino program using Arduino IDE.

Step 9: Pair the Bluetooth modules and verify communication.

Step 10: Test each voice command individually and ensure correct vehicle movement.

12. Testing & Validation

12.1 UART Communication Test using RealTerm

Objective: To verify that the VC-02 module transmits the correct hexadecimal commands.

Procedure:

- Connected the VC-02 module to a USB-to-TTL converter.
- Opened the RealTerm serial terminal.
- Spoke each predefined command.
- Observed the hexadecimal values displayed on the screen.

Observation:

The hexadecimal values corresponding to Forward, Backward, Left, Right and Stop were displayed correctly.

Result:

The VC-02 module successfully generated and transmitted the expected hexadecimal

commands.

12.2 Bluetooth Communication Test

Objective: To verify wireless communication between HC-05 and HC-06.

Procedure:

- Connected HC-06 to the Arduino Uno.
- Paired HC-05 with HC-06.
- Opened the Arduino Serial Monitor.
- Sent voice commands using the remote.

Observation:

The received hexadecimal values were displayed correctly on the Serial Monitor.

Result:

Successful Bluetooth communication was established.

12.3 Final Vehicle Test

Objective: To verify the complete working of the robotic vehicle.

Procedure:

- Powered the remote and vehicle.
- Established Bluetooth communication.
- Issued all five voice commands.

Observation:

The robot correctly executed Forward, Backward, Left, Right and Stop commands.

Result:

The Voice Controlled Toy Cybertruck functioned successfully during demonstration.

13. Applications

Educational robotics projects.
Embedded systems learning.
Robotics exhibitions and demonstrations.
Voice-controlled automation experiments.
Student mini-projects.

14. Advantages

Offline voice recognition.
Wireless Bluetooth communication.
Simple and user-friendly operation.
Rechargeable battery powered.
Portable design.
Custom Cybertruck-inspired body.

15. Limitations

Supports only predefined voice commands.
Bluetooth range is limited.
No obstacle avoidance feature.
Requires charged batteries for reliable operation.

16. Bill of Materials

Component	Quantity	Approx. Cost (₹)	Purpose
Arduino Uno	1	700	Main Controller
L293D Shield	1	350	Motor Driver
HC-05	1	350	Bluetooth Transmitter
HC-06	1	300	Bluetooth Receiver
VC-02	1	480	Voice Recognition
BO Motors	4	600	Movement
Li-ion Cells	3	900	Power Supply
Robot Chassis	1	150	Base
Rocker Switch	2	60	Power Control
Cardboard	2	100	Body

17. References

https://youtu.be/aE_J7B-O4VQ?feature=shared

<https://youtu.be/P0UDOAZBqr4?si=rSIOShRFMLGXxSN0>

https://drive.google.com/drive/folders/1PpOmFiRZvzJ4zT9uqntbI_LMkhfzL4Tj?usp=sharing